Dual-Listing Spread Dynamics

Institutional Arbitrage Mechanisms and the Law of One Price
in Chinese Cross-Listed Equities, 2014–2026

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https://github.com/rishabh-verma-stat/dual-listing-spread-dynamics.git

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Executive Summary

The same Chinese company can be listed in Shanghai and Hong Kong (an A–H pair), or in Hong Kong and New York (an H–ADR pair). Both pairs represent identical claims on identical cash flows, so under the law of one price both should trade at parity. They do not behave the same way at all.

Over 2014–2026, using identical code, identical tests and identical multiple-testing corrections on both groups:

	A–H (63 pairs)	**H–ADR** (20 pairs)
Mean premium	+77.95%	−0.15%
Mean standard deviation	≈ 40%	2.95%
ADF rejects unit root	0 / 63	20 / 20
Engle–Granger cointegrated	0 / 63	20 / 20
Mean cointegrating β	0.770 (spurious)	**0.994**
Mean R^2	≈ 0.70 (spurious)	**0.9927**
Median AR(1) ϕ	0.9936	0.0972
Half-life of a shock	≈ 107 days	< 1 day, unresolvable
Conversion mechanism	**none**	**deposit agreement**

The two groups differ by two orders of magnitude in both the level and the dispersion of the spread. The economics are identical; what differs is plumbing. H–ADR pairs have a depositary bank contractually obliged to convert k Hong Kong shares into one receipt and back on demand. A–H pairs have no such mechanism, and mainland capital controls prevent the trade being replicated synthetically.

The law of one price is enforced by institutional machinery, not by markets. Where the machinery exists, prices are cointegrated with $\beta \approx 1$ and $R^2 > 0.99$. Where it does not, a 78% gap survives twelve years of scrutiny by every participant in two of the world's largest equity markets.

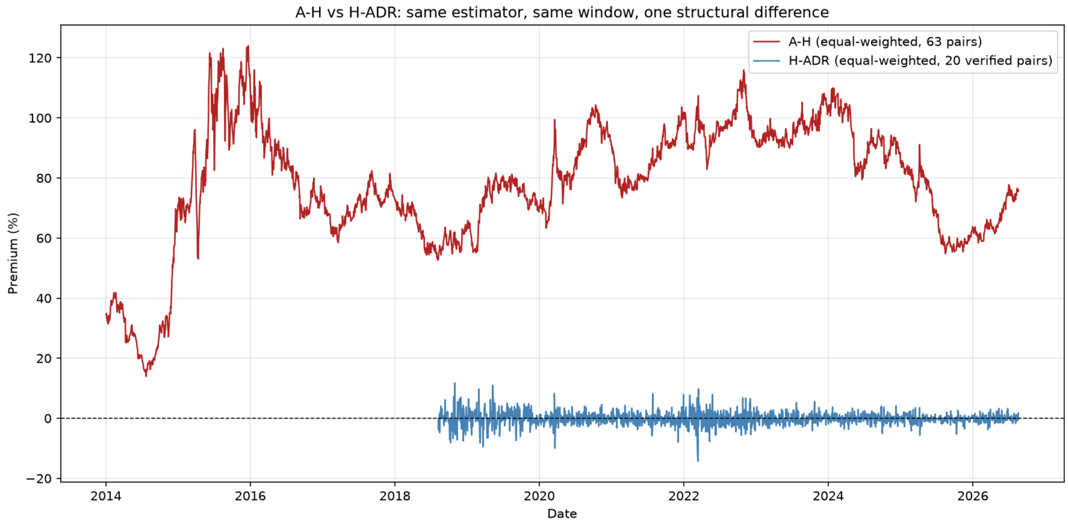

Figure 1: Equal-weighted A-H premium (63 pairs) versus H-ADR premium (20 verified pairs), same estimator and sample window. The scale difference is itself the paper's central finding.

1 Introduction

1.1 The Law of One Price and the Role of Arbitrage

Economics has few laws as clean as physics, but the law of one price comes close in a sense that it is the best nominee for the title of second law of economics (the first law of economics being the law of supply and demand, and a fine law it is): identical assets, once expressed in common units, should command identical prices (Lamont and Thaler, 2003). The mechanism enforcing it is not regulatory but private. If two claims to the same cash flows trade at different prices, a trader can buy the cheap one and sell the expensive one, earning a profit that requires no view on the underlying asset at all — only that the gap will close. Pursuing that profit is what closes the gap. No investor needs to be rational for the law to hold; it is enough that sufficient capital is willing to act on the discrepancy.

The qualification is that arbitrage requires a mechanism: some means of converting the cheap claim into the expensive one, or of shorting one leg against the other. Where that mechanism is missing — through capital controls, short-sale restrictions, or the simple absence of a conversion channel between two markets — a price gap can persist not because investors have failed to notice it, but because noticing it confers no way to act on it. Lamont and Thaler illustrate this with the Infosys ADR, which traded at a 136% premium to its Bombay-listed shares in 2000 because capital controls prevented American investors from buying the cheaper local shares and creating additional ADRs against them (Lamont and Thaler, 2003).

This distinction — between a price gap that persists because it cannot be arbitraged, and one that persists despite arbitrage being available — is the organizing question of this paper. Chinese dual-listed equities provide an unusually clean setting to study it, because the same set of underlying companies is dual-listed under two different institutional regimes: one where a conversion mechanism exists, and one where it does not.

1.2 The A–H Premium

Chinese dual-listed equities are the largest and most persistent instance of this phenomenon in global markets. By 2024, some 150 mainland firms carried both an A-share listing (Shanghai or Shenzhen) and an H-share listing (Hong Kong), representing a combined market capitalization exceeding 29 trillion RMB — more than a third of mainland China's total equity market (Liu and Zou, 2026). The two share classes confer identical cash-flow and voting rights, yet A-shares have traded at a premium to their H-share twin in the overwhelming majority of firm-months since the early 2000s, with the capitalization-weighted Hang Seng AH Premium Index averaging roughly 27% and ranging between 3% and 50% (Zhang and Zhang, 2025). Unlike Royal Dutch/Shell or other developed-market Siamese-twin pairs, whose deviations rarely exceed low double digits and have narrowed over time (Froot and Dabora, 1999), the A–H premium has remained large and volatile across two decades, multiple regulatory regimes, and the introduction of formal cross-border trading channels. That last point is the puzzle this paper is built around. The Shanghai–Hong Kong Stock Connect (November 2014) and Shenzhen–Hong Kong Stock Connect (December 2016) were designed explicitly to narrow this gap, opening quota-limited northbound and southbound channels between the two markets. The premium did not narrow. In the sample studied here, the equal-weighted premium roughly doubled in the year following the Shanghai launch. This is not an isolated observation: Liu and Zou (2026) document the same pattern and argue that Stock Connect widened rather than disciplined the premium, because it expanded access primarily for retail investors, who are more prone to the behavioral preferences that sustain it. If integration through retail participation can widen a mispricing that integration through institutional arbitrage would be expected to close, the presence or absence of a working arbitrage *mechanism* — as distinct from market access in general — becomes the central question.

1.3 This Paper

A large literature explains the A–H premium through market segmentation (Sun and Tong, 2000; Wang and Jiang, 2004), informational asymmetry (Fernald and Rogers, 2002), differences in investor composition and risk preferences (Mei et al., 2005), and, more recently, behavioral valuation gaps (Liu and Zou, 2026) and extrapolative price expectations (Zhang and Zhang, 2025). These studies are informative about *why* A-shares are more attractive to their marginal investor. Almost none of them, however, study the A–H market against a matched control group in which a conversion mechanism does exist. Without such a comparison, it is difficult to separate two distinct claims: that Chinese equity markets are unusually prone to mispricing, and that a specific institutional channel — the absence of any instrument converting an A-share into an H-share — is what allows a mispricing to survive undisciplined for over a decade. This paper constructs that comparison directly. Alongside 63 A–H pairs, it studies 20 Hong Kong–ADR pairs: the same country, a similar set of large, liquid firms, and an almost identical accounting identity for the premium, but with one structural difference. An ADR is backed by a fixed number of ordinary shares held in custody, and a depositary bank is contractually obliged to create and cancel ADRs against those shares on demand (Lamont and Thaler, 2003). Running an identical battery of stationarity and cointegration tests on both groups isolates the effect of that one difference. The result, previewed in the Executive Summary, is stated most simply as a contrast: A–H prices show no evidence of cointegration under any test applied here, while H–ADR prices are cointegrated with an estimated coefficient within a percentage point of one for every pair studied. The gap is not explained by the underlying economics being different in Hong Kong versus New York; it is explained by whether a bank is obliged to close it.

2 Institutional Background

2.1 Share classes

A-shares raise CNY from mainland savers; H-shares raise HKD and USD from international institutions. The two listings tap separate, non-overlapping pools of capital: the mainland pool is enormous, sustained by a high domestic savings rate and a narrow menu of investment outlets, but legally sealed off from foreign capital; the Hong Kong pool is precisely the international capital that the mainland pool cannot reach. A company that lists twice accesses both rather than choosing between them.

This does not make an A-share "the same stock on a different exchange," in the way a US Treasury bond trades identically wherever it is quoted. A-shares and H-shares are legally distinct instruments, with distinct **ISIN** codes, despite being claims on the same firm. No clearing system can convert one into the other, because there is no shared identity for the systems to match against. The contrast with an ADR is instructive: an ADR carries a formal, contractual link back to the ordinary share it represents — the deposit agreement discussed in Section 2.3 — and it is precisely that link which is absent here. Even if the Hong Kong market prices H-shares efficiently on its own terms, that alone cannot close the A–H gap, because closing a gap between two markets requires an instrument connecting them, not merely calm pricing within each.

2.2 Why the A–H gap cannot be arbitraged

The barrier is regulatory rather than structural. China restricts capital flows across its border — not information, but money — so a mainland investor historically could not simply move CNY to Hong Kong and buy H-shares directly, nor could foreign capital flow freely into A-shares. This alone would be enough to sustain a persistent gap: even a fully rational, well-capitalised trader who correctly identifies the mispricing has no legal channel to act on it.

Two further frictions compound the problem, and either alone would be sufficient to block the trade even where capital could cross the border. First, there is no conversion instrument between A- and H-shares in either direction — unlike an ADR, no institution stands ready to turn one into the other on demand (Section 2.3). Second, short-selling A-shares is heavily restricted, so even an investor who could access the mainland market cannot easily construct the short leg of a convergence trade. The combination — no cross-border capital flow, no conversion mechanism, and no reliable short — means the arbitrage cannot be executed even synthetically, regardless of how confident a trader is that the gap will eventually close.

The Shanghai–Hong Kong and Shenzhen–Hong Kong Stock Connect programs (2014, 2016) are the partial exception to the capital-control barrier: a quota-limited, government-administered channel permitting a restricted volume of cross-border trading. Their effect on the premium is discussed in Section 2.4.

2.3 Why the H–ADR gap can be arbitraged

The deposit agreement between the issuer, the depositary bank and ADR holders obliges the bank to perform creation (k Hong Kong shares $\rightarrow$ 1 ADR) and cancellation (1 ADR $\rightarrow$ k Hong Kong shares) on demand, for a small conversion fee. This is a contractual, court-enforceable obligation, not a market convention that could break down under stress — the "pipe" whose presence or absence this paper tests.

The mechanism directly enforces the law of one price. If the ADR trades above the value of k Hong Kong shares, a trader can buy k shares in Hong Kong, deliver them to the depositary bank for conversion, and sell the resulting ADR in New York at the higher price — a round trip requiring no view on the underlying company, only that the bank will honour its contractual obligation. The reverse trade runs when the ADR trades below parity. Because the conversion ratio k is fixed by contract, this is a considerably lower-risk trade than betting on an A–H gap eventually closing: the two legs are the same claim rather than merely correlated claims, and the mechanism connecting them is legal rather than merely economic.

2.4 Stock Connect

Shanghai–Hong Kong Stock Connect (November 2014) and Shenzhen–Hong Kong Stock Connect (December 2016) introduced quota-limited northbound and southbound channels, permitting Hong Kong investors to trade eligible A-shares and mainland investors to trade eligible H-shares within regulator-set limits.

Note the puzzle: the naive prediction is that opening a channel between two segmented markets should narrow the price gap between them, as arbitrage capital flows through it. The A–H premium did the opposite. In the sample studied here, the equal-weighted premium rose from 52.1% in 2014 to 127.1% in 2015, in the immediate aftermath of the Shanghai launch. Zhang et al. (2022) document the same pattern and offer an explanation: Stock Connect expanded market *access* primarily for retail investors, and retail investors are the population whose behavioural preferences — loss aversion, probability weighting, a taste for lottery-like payoffs — already made A-shares more attractive to their marginal buyer. Zhang et al. (2022), studying the policy's effect on the premium directly, similarly find no straightforward reduction but rather it promoted the premium. Integration that expands participation by investors predisposed toward the expensive side of a mispricing widens it rather than disciplining it.

This distinction — between market *access* and an arbitrage *mechanism* — is the paper's organising question restated at the institutional level. Stock Connect gave more investors the ability to place an order in the other market; it did not give any investor a contractual instrument for converting one share class into the other, nor did it relax the restrictions on shorting A-shares. Access without mechanism, on this account, is not equivalent to arbitrage.

2.5 Price limits and state intervention

Two further features of the mainland market shape the premium's dynamics independently of capital controls.

Daily price limits. A-shares cannot move more than ±10% in a single session. During periods of large H-share moves, Shanghai can therefore be mechanically unable to track Hong Kong, regardless of investor willingness to trade. The data quality guards applied in Section 4 identify a clean example: on 2 September 2015, the Shanghai leg of ICBC's A–H pair rose by exactly 10.00% (2.856737/2.597034 = 1.09999...), the daily limit binding precisely, while the Hong Kong leg fell 1.3% over the same session. The premium widened by roughly 16 percentage points in two trading days — not because Shanghai investors were unwilling to bid the price higher, but because regulation prevented it.

State intervention. A separate and unrelated episode in early July 2015 shows a different mechanism. Over 3–8 July 2015, ICBC's A-share leg rose sharply (up to +9.04% in a single session, short of the 10% limit) while the H-share leg fell, producing genuine divergence rather than a limit-constrained one. This coincides with the period in which China's "national team" of state-backed funds intervened directly in the market to arrest the 2015 (Li et al., 2024). The purchases were concentrated in index-heavyweight, state-owned firms — of which ICBC is among the largest — and Li and Liu (2024) find that National Team ownership was associated with *increased*, not decreased, stock return volatility after 2015, consistent with a purchase programme that shifted prices rather than stabilised them. Consistent with this, the same divergence is present in other large state-owned banks over the same window but absent in privately-held firms in the sample such as BYD and Great Wall Motor, though this comparison is suggestive rather than a formal test.

3 Literature

3.1 Segmentation and frictions

The earliest explanations for A-H mispricing treat the two markets as genuinely separate pools of investors, priced by different discount rates rather than by a shared valuation of the same cash flows. Fernald and Rogers (2002) formalise this within a standard dividend-discount model: given that domestic investors historically paid roughly four times what foreign investors paid for the same claim, the implied gap in required returns between the two investor bases is only around four percentage points — a modest and economically plausible figure once one accounts for the limited investment alternatives available to mainland savers. On this account, the premium is not evidence of irrationality on either side; it is what a standard asset-pricing model predicts once two investor populations face genuinely different opportunity sets.

Wang and Jiang (2004) provide direct evidence for the segmentation this requires. Studying sixteen A-H pairs, they find H-shares carry substantially higher exposure to Hong Kong market factors than to the mainland market, while A-shares show the reverse: H-shares behave, in their words, more like Hong Kong stocks than mainland Chinese stocks. If the two share classes were priced by a common, integrated pool of capital, no such asymmetry would be expected. They also find little Granger-causality in either direction between A- and H-share returns, which is difficult to reconcile with the two markets sharing a single information-processing mechanism, and consistent instead with two markets pricing largely independently of one another.

A related literature documents that this segmentation produces the *opposite* pattern from most other markets with restricted and unrestricted share classes. Sun and Tong (2000) note that where foreign investors are typically charged a premium for access to a segmented market, China's B-shares — available only to foreign investors — historically traded at a *discount* to their domestic A-share counterparts. Their explanation is a substitution argument: H-shares and Hong Kong "red-chip" stocks provide foreign investors with a close substitute for B-shares, making foreign demand for B-shares more elastic and widening the discount. The finding is a reminder that segmentation alone does not pin down the *sign* of a cross-listing gap; the direction depends on which side of the market has more concentrated, less substitutable demand, and in China's case this has consistently pushed A-shares' domestic listing above its foreign-accessible twin, whichever form that twin takes.

Together, these studies establish an important boundary condition for this paper. Segmentation, on its own, is a description of the premium's existence, not of its persistence: nothing in this literature explains why a bank of arbitrage capital, once it identifies a mispricing this large, cannot simply close it. That question — not whether the two markets are segmented, but whether the segmentation is enforced by anything an arbitrageur cannot get around — is the subject of Section 2.2.

3.2 Behavioural explanations

A separate strand of the literature accepts that segmentation exists but asks why arbitrageurs, informed and unconstrained by law in principle, do not simply trade the gap away. Three recent studies propose distinct mechanisms, and read together they triangulate on the same conclusion from different directions.

Mei et al. (2005) locate the mechanism in speculative *trading* itself. Using 75 A-B pairs from 1993–2000, in a market with strict short-sale constraints, they show that A-share turnover is a strong and robust predictor of the cross-sectional premium, explaining roughly 20% of its monthly variation even after controlling for liquidity, risk, and discount-rate proxies. Their interpretation, following the resale-option models of Harrison and Kreps (1978); Scheinkman and Xiong (2003), is that when investors cannot short an overvalued share, the marginal buyer is always an optimist who additionally values the option to resell to a still-more-optimistic buyer later; the premium is partly the price of that option, and it should — and does — rise with turnover.

Liu and Zou (2026) instead locate the mechanism in *investor psychology*, independent of how often a share changes hands. Drawing on prospect theory, they argue that retail investors — who dominate trading in A-shares far more than in H-shares — systematically overweight the potential for large gains and are more averse to losses than a standard rational investor. Applied to a share's own history of past returns, this generates a subjective "attractiveness score" that runs higher for A-shares than for the matched H-share of the same company, even though both promise identical future cash flows. Across the full 2000–2024 dual-listed universe, this attractiveness gap is a strong and robust predictor of the premium, surviving a wide range of controls and alternative specifications. The result complements Mei et al. (2005): where turnover captures a speculative *trading* motive, this finding suggests that even a buy-and-hold retail investor may value the same cash flows differently on the two listings, simply because of how gains and losses are weighted.

Zhang and Zhang (2025) propose a third, more minimal mechanism: extrapolative price *expectations*, requiring neither short-sale constraints nor prospect-theoretic preferences. Investors simply forecast future capital gains by extrapolating recent past returns; if A-shares happen to rise faster than H-shares for any initial reason, that divergence becomes partially self-fulfilling as higher realised gains feed higher expected gains. Calibrated to the post-Connect period, their model reproduces the mean, volatility, and persistence of the Hang Seng AH Premium Index — which itself averages 127 (i.e. a 27% premium) against a theoretical value of 100 — using a small, economically plausible difference in the rate at which investors extrapolate A-share versus H-share returns.

These three mechanisms are not mutually exclusive, and none of them requires the absence of an arbitrage channel to operate — all three authors explicitly note that the AH-market has, since 2014, been formally connected by Stock Connect. Zhang and Zhang (2025) show directly that convergence trading in this setting is unattractive rather than impossible: a simulated one-year convergence trade earns a positive mean return but carries a probability of loss exceeding 30% and pronounced negative skewness, so that even a fully rational arbitrageur with correct beliefs may reasonably decline the trade. This is an important distinction for the present paper. The behavioural literature explains why an *available* arbitrage channel goes underused; it is a separate question, addressed in Section 2.2, why for A-H pairs no such channel exists to be used or declined in the first place.

3.3 Fundamentals: dividends and currency

A smaller but directly relevant literature asks not why the premium exists, but what moves it at the margin. Jiao et al. (2022) study a panel of A-H pairs from 2006–2019 and find the premium is systematically *smaller* for companies that pay regular cash dividends, and smaller again when the renminbi is expected to appreciate against the Hong Kong dollar. Their explanation, formalised in a discounted-dividend model, is that international H-share investors place a premium on cash they can actually extract from the company and convert into their own currency: a dividend-paying, currency-favourable H-share draws in more demand from its international buyer base, which narrows the gap rather than widening it.

This finding is directly relevant to a methodological choice made in this paper. Section 4.5 documents that switching from dividend-adjusted to unadjusted closing prices moves the mean A-H premium by roughly seventeen percentage points, which we attribute to the two share classes being adjusted for dividends inconsistently by the data vendor. Jiao et al. (2022) provide independent evidence that dividend treatment is a first-order driver of the measured premium, not a second-order nuisance — which is precisely why getting it right, rather than leaving it to a vendor's default assumption, was worth the trouble.

3.4 Cointegration studies

Two papers speak most directly to the question this paper revisits. Cai et al. (2011) apply a Markov-switching error-correction model to A-H prices from January 1999 to March 2009 and find that cointegration strengthened over the decade, attributing the improvement to capital account liberalisation and declining information asymmetry. Pan et al. (2012) test 44 dual-listed firms on minute-level intraday data from July 2007 to February 2009 and report that "almost all" A and H prices are cointegrated.

A closer reading of Pan et al. (2012)'s own results, however, shows this summary understates considerable internal inconsistency. Of the subset reported in their Table 5, several firms cointegrate on some intraday windows but not others — ICBC, this paper's own test case, cointegrates on only two of six reported windows — and a handful fail to cointegrate on any window at all. Their own arbitrage simulation, designed to test whether the estimated relationship was tradeable, found close to a coin flip's worth of positive-profit days and concluded explicitly that "the arbitrage simulation did not result in a short term co integration of the two equity markets." Read this way, the earlier literature does not report a clean, stable, tradeable cointegrating relationship so much as a fragile one, sensitive to sampling window and firm choice, on a sample separated from the present one by less than two decades.

This paper's finding — no A-H pair cointegrates after MacKinnon critical values and FDR correction, on 2014–2026 data — can therefore be read two ways. It may reflect a genuine change in market structure around Stock Connect, as Zhang et al. (2022) and Tang and Liu (2026) suggest the premium itself did. Alternatively, it may reflect that the earlier literature's cointegration findings were themselves closer to the noise floor than commonly summarised, and a stricter multiple-testing correction applied to their own data might have produced a similar result. Both readings point to the same methodological conclusion: cointegration claims on A-H prices are highly sensitive to sample period, firm selection, and statistical correction, and should be reported with the uncertainty this implies rather than as a settled stylised fact.

3.5 The law of one price beyond China

It is worth stating plainly what this paper's design tests, because the intuitive prior is not obvious. One might reasonably expect the opposite of the result reported here: that a fully open, unrestricted market, with no institutional force tying two independently-traded claims together, would behave close to a random walk relative to each other, while a market segmented by an explicit, persistent regulatory restriction — itself a consistent, rule-governed constraint — might be expected to generate *some* detectable statistical structure in the price relationship, even a loose or imperfect one. The finding reported in Sections 7 and 3.4 is the reverse: the unrestricted H–ADR pairs are tightly cointegrated, while the restricted A–H pairs show no cointegrating relationship under any test applied here, despite the same capital-control regime having been continuously in place for over a decade. A restriction, it turns out, does not by itself manufacture a detectable relationship; what generates one is a specific, engineered bridge between the two markets — the depositary conversion mechanism — and its absence, not the mere presence of a restriction, appears to be what matters.

Froot and Dabora (1999) provide an important further data point on this question, and complicate it usefully. They study three "Siamese-twin" pairs — Royal Dutch/Shell, Unilever N.V./PLC, and SmithKline Beecham — whose corporate charters fix an exact division of current and future cash flows between two independently traded stocks, with no capital controls, no restrictions on shorting either leg, and full liquidity in both listings: by any measure, a genuinely unrestricted setting. Using sixteen years of daily data, they find the price differential between each pair is strongly correlated with the stock market index and currency of whichever country each twin trades most actively in, despite the identical underlying cash flows, and augmented Dickey–Fuller tests fail to reject a unit root in the price differential for all three pairs — the authors state directly that they "cannot reject the hypothesis that the half-life is infinite." An unrestricted market, in their setting, did *not* straightforwardly produce tight cointegration either.

Read together with the present paper's results, the pattern that emerges is not a simple binary of restricted-versus-free markets. It is closer to Froot and Dabora's own conclusion: "lack of disciplinary arbitrage does not explain why there are deviations in the first place." What appears to matter is not whether a market is *restricted*, nor even whether arbitrage is *legally available*, but whether a specific, low-friction, close-to-automatic conversion instrument exists — an ADR's depositary mechanism obliges conversion on demand, with no directional capital commitment and no view required on how long a gap will persist, while even a fully legal, fully liquid twin-share arbitrage still requires an investor to hold a basis position of uncertain duration, exposed to the agency and benchmarking frictions Froot and Dabora identify. The absence of restriction is not sufficient for parity; the absence of a conversion bridge appears, in this paper's setting, to be sufficient for its failure.

4 Data and Methodology

4.1 Sample construction

A–H universe: the 56 cross-listed firms catalogued in Pan et al. (2012), extended with post-2012 dual listings; 66 attempted, **63 retained** (3 delisted). H–ADR universe: sponsored, exchange-listed dual listings only; 26 attempted, **20 retained** after ratio verification.

Unsponsored OTC ADRs are excluded: absent a deposit agreement guaranteeing two-way conversion, the mechanism under study does not exist and their pricing carries no information about it.

Survivorship. 3 of 66 A–H pairs are no longer listed. The sample is conditioned on survival.

4.2 The premium

For A–H, with e_t the HKD-per-CNY rate:

$$\text{premium}_t^{AH} = \frac{P_t^A\, e_t}{P_t^H} - 1.$$

For H–ADR, with k the conversion ratio and f_t the HKD-per-USD rate:

$$\text{premium}_t^{ADR} = \frac{P_t^{ADR}\, f_t}{k\, P_t^{HK}} - 1.$$

4.3 Estimating the conversion ratio

k is set by contract but not observable in price data. A wrong k manufactures a premium out of arithmetic alone:

$$\frac{P^{ADR} f}{k_{\text{wrong}} P^{HK}} - 1 \;\approx\; \frac{k_{\text{true}}}{k_{\text{wrong}}} - 1.$$

We therefore estimate it, exploiting the fact that k is a known integer:

$$\hat{k} = \text{median}_t\left(\frac{P_t^{ADR} f_t}{P_t^{HK}}\right), \qquad k^* = \text{round}(\hat{k}).$$

Two diagnostics, both of which must pass at the 1% level:

$$\varepsilon = \frac{|\hat{k} - k^*|}{k^*} \quad \text{(wrong integer)}, \qquad \delta = \frac{|\hat{k}_{\text{2nd half}} - \hat{k}_{\text{1st half}}|}{\hat{k}_{\text{1st half}}} \quad \text{(ratio changed)}.$$

The integrality of k turns a nuisance parameter into a specification test: if the two tickers are not the same claim, $\hat{k}$ does not land on an integer.

Table 1: Conversion ratio verification for the 25 H–ADR candidates. $\hat{k}$ is the median daily implied ratio; k^* is the nearest integer; $\varepsilon = |\hat{k} - k^*|/k^*$ tests whether the ratio is a clean integer; δ tests stability between the first and second half of the sample. A pair is verified only if both $\varepsilon < 1\%$ and $\delta < 1\%$.

Company	$\hat{k}$	k^*	ε	δ	Verified
New Oriental Ed.	9.987	10	0.13%	0.07%	Yes
NetEase	4.993	5	0.14%	0.16%	Yes
JD.com	1.997	2	0.16%	0.34%	Yes
BeiGene	12.977	13	0.18%	0.24%	Yes
Zai Lab	9.982	10	0.18%	0.21%	Yes
ZTO Express	0.998	1	0.19%	0.12%	Yes
Baozun	3.006	3	0.22%	0.19%	Yes
Alibaba	7.981	8	0.23%	0.38%	Yes
Yum China	0.998	1	0.24%	0.17%	Yes
Qifu Technology	1.995	2	0.24%	0.80%	Yes
Autohome	4.010	4	0.24%	0.51%	Yes
Baidu	7.980	8	0.24%	0.29%	Yes
XPeng	1.994	2	0.28%	0.48%	Yes
Trip.com	0.997	1	0.29%	0.31%	Yes
Li Auto	1.994	2	0.30%	0.09%	Yes
Miniso	3.986	4	0.34%	0.16%	Yes
Bilibili	0.996	1	0.38%	0.14%	Yes
H World (Huazhu)	9.952	10	0.48%	0.20%	Yes
Weibo	0.995	1	0.49%	0.14%	Yes
NIO	0.992	1	0.75%	0.42%	Yes
Tuya	0.992	1	0.81%	1.82%	No
GigaCloud	23.769	24	0.96%	80.40%	No
Kanzhun (BOSS)	2.023	2	1.17%	1.16%	No
Hutchmed	4.935	5	1.30%	0.48%	No
Lufax	1.945	2	2.74%	0.94%	No

4.4 Sensitivity to the exclusion threshold

Threshold	Pairs	Mean premium	Mean s.d.
0.5%	17	−0.16%	2.88%
1.0%	20	−0.15%	2.95%
2.0%	23	−0.18%	3.19%
5.0%	24	−0.40%	3.75%
no filter	25	+1.13%	7.90%

A 200-fold change in the threshold moves the estimate by ≈ 1.3pp.

4.5 Adjusted versus unadjusted prices

Vendor "adjusted" prices apply dividend and split factors that may differ between two listings of the same security. For H–ADR pairs this produced a systematic wedge: the implied ratio was biased downward in the first half of the sample for *all ten* affected pairs. Switching to unadjusted closes collapsed the drift and doubled the verified sample from 10 to 20.

For A–H the same switch lowered the mean premium from +94.92% to +77.95%. This gap plausibly reflects the different dividend treatment of the two share classes: A-shares and H-shares of the same company can differ in dividend timing and withholding-tax treatment (mainland versus international holders), and Jiao et al. (2022) document that cash dividends are a significant, negative predictor of the A–H premium in their own panel. If a data vendor's adjustment procedure strips these dividends inconsistently across the two listings — as it evidently does for H–ADR pairs, per the preceding discussion — the resulting artefact would plausibly move in the same direction observed here. **All qualitative conclusions hold under either choice**; 63/63 pairs remain positive.

Trade-off: unadjusted prices retain ex-dividend drops, adding genuine one-day jumps to the series.

4.6 Data quality guards

Stale-price detection (runs of ≥ 3 identical closes; A-shares are frequently suspended and vendors forward-fill) and scale-free jump detection ($|\Delta \log(1 + \text{premium})| > 0.10$). Flagged episodes are investigated, not deleted.

Table 2: Staleness and jump diagnostics by A-H pair. `stale_a`/`stale_h` give the count of trading days with zero price change (a proxy for illiquid/stale quotes) on the A- and H-share legs respectively; `jumps` counts log-return outliers flagged at the $|z| > 5$ threshold; n is the number of aligned trading-day observations; `stale_pct` is $\max(\text{stale\_a}, \text{stale\_h})/n \times 100$.

Name	stale_a	stale_h	jumps	n	stale_pct (%)
Chongqing Iron & Steel	428	176	40	2977	10.14
Guangzhou Shipyard	266	142	25	2977	6.85
Luoyang Glass	249	134	48	2977	6.43
Yizheng Chemical	143	216	26	2977	6.03
China Shipping Container	126	173	14	2977	5.02
Beiren Printing	131	157	49	2977	4.84
COSCO Shipping	144	81	19	2977	3.78
China Shipping Dev	93	84	2	2977	2.97
Beijing North Star	29	124	7	2977	2.57
CRRC	80	73	12	2977	2.57
Nanjing Panda	76	63	33	2977	2.33
ZTE	74	61	13	2977	2.27
Chalco	108	10	4	2977	1.98
Shandong Xinhua Pharma	61	56	21	2977	1.97
China Railway Group	96	12	9	2977	1.81
Guangzhou Pharmaceutical	69	27	5	2977	1.61
Dongfang Electric	75	0	14	2977	1.26
Bank of China	52	22	4	2977	1.24
China Shenhua	73	0	12	2977	1.23
Zijin Mining	34	26	4	2977	1.01
Datang Power	14	46	8	2977	1.01
New China Life	25	25	4	2977	0.84
Shanghai Petrochemical	9	39	17	2977	0.81
China Eastern Airlines	29	19	12	2977	0.81

continued on next page

Table 2 – continued

Name	stale_a	stale_h	jumps	n	stale_pct (%)
Guangshen Railway	16	27	1	2977	0.72
Air China	20	20	6	2977	0.67
Great Wall Motor	20	14	13	2977	0.57
Tianjin Capital Env.	5	26	13	2977	0.52
Maanshan Iron	11	14	12	2977	0.42
Minsheng Bank	18	6	2	2977	0.40
Bank of Communications	18	6	3	2977	0.40
CITIC Bank	22	0	3	2977	0.37
Huadian Power	13	3	8	2977	0.27
Sinopec	9	7	3	2977	0.27
Chenming Paper	8	7	10	2977	0.25
China Railway Const.	9	6	8	2977	0.25
Angang Steel	6	9	6	2977	0.25
BYD	7	7	3	2977	0.24
Anhui Expressway	8	6	15	2977	0.24
CITIC Securities	14	0	8	2977	0.24
China Coal Energy	0	12	6	2977	0.20
Anhui Conch Cement	6	6	2	2977	0.20
Guangzhou Automobile	10	0	8	2977	0.17
Huaneng Power	3	7	5	2977	0.17
PetroChina	9	0	2	2977	0.15
Jiangxi Copper	8	0	5	2977	0.13
China Merchants Bank	3	3	1	2977	0.10
Shenzhen Expressway	0	6	4	2977	0.10
China Pacific Insurance	3	0	3	2977	0.05
China Southern Airlines	3	0	13	2977	0.05
China Comm. Construction	0	3	12	2977	0.05
ICBC	0	3	2	2977	0.05
Huatai Securities	0	0	4	2644	0.00
Tsingtao Brewery	0	0	3	2977	0.00
Guangdong Kelon	0	0	16	2977	0.00
Yanzhou Coal	0	0	21	2977	0.00
China Construction Bank	0	0	4	2977	0.00
Jiangsu Expressway	0	0	2	2977	0.00
China Life	0	0	5	2977	0.00
Weichai Power	0	0	3	2977	0.00
China Oilfield Services	0	0	5	2977	0.00
Ping An	0	0	1	2977	0.00
CNOOC	0	0	0	1019	0.00

5 Descriptive Results

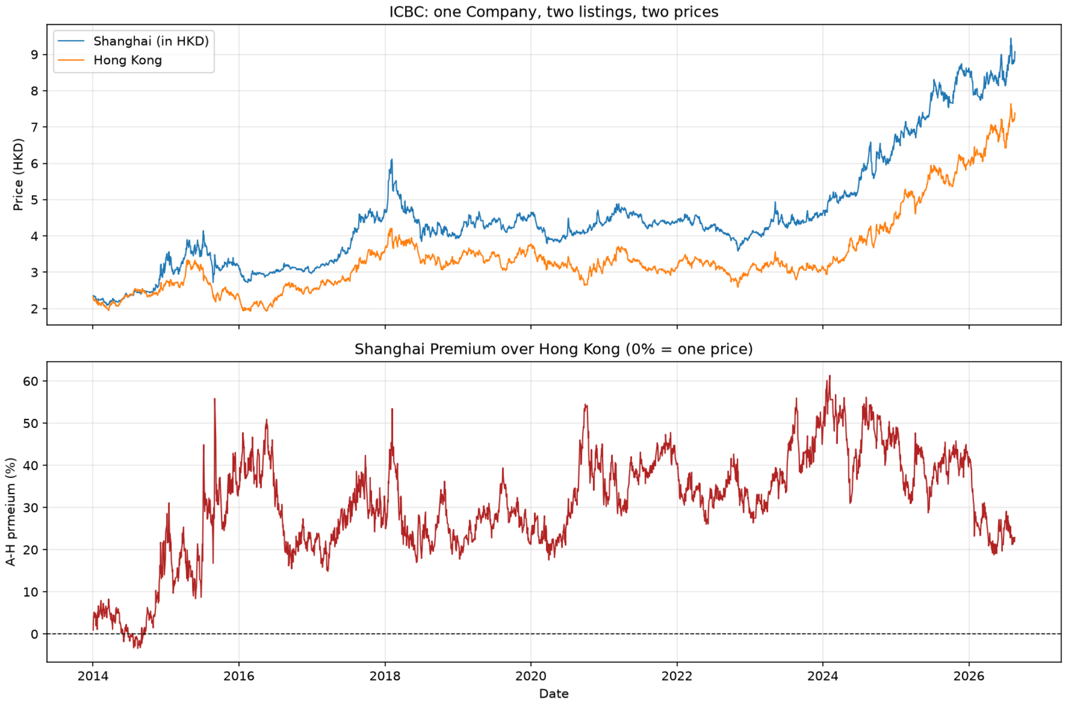

Figure 2: ICBC: Shanghai and Hong Kong prices (top) and the A-H premium (bottom), 2014–2026.

Key numbers: 63/63 A–H pairs positive; range +3.54% (China Merchants Bank) to +327.08% (Luoyang Glass); mean of means +77.95%, median +63.40%; cross-sectional skew +1.37.

Reconciliation with published indices. The Hang Seng AH Premium Index averages ≈ 27% over comparable periods (Zhang and Zhang, 2025), against +77.95% here. Both are correct: the index is capitalisation-weighted and therefore dominated by the megacap banks that populate the bottom of our cross-section, whereas an equal-weighted mean describes the *typical pair*. We report both.

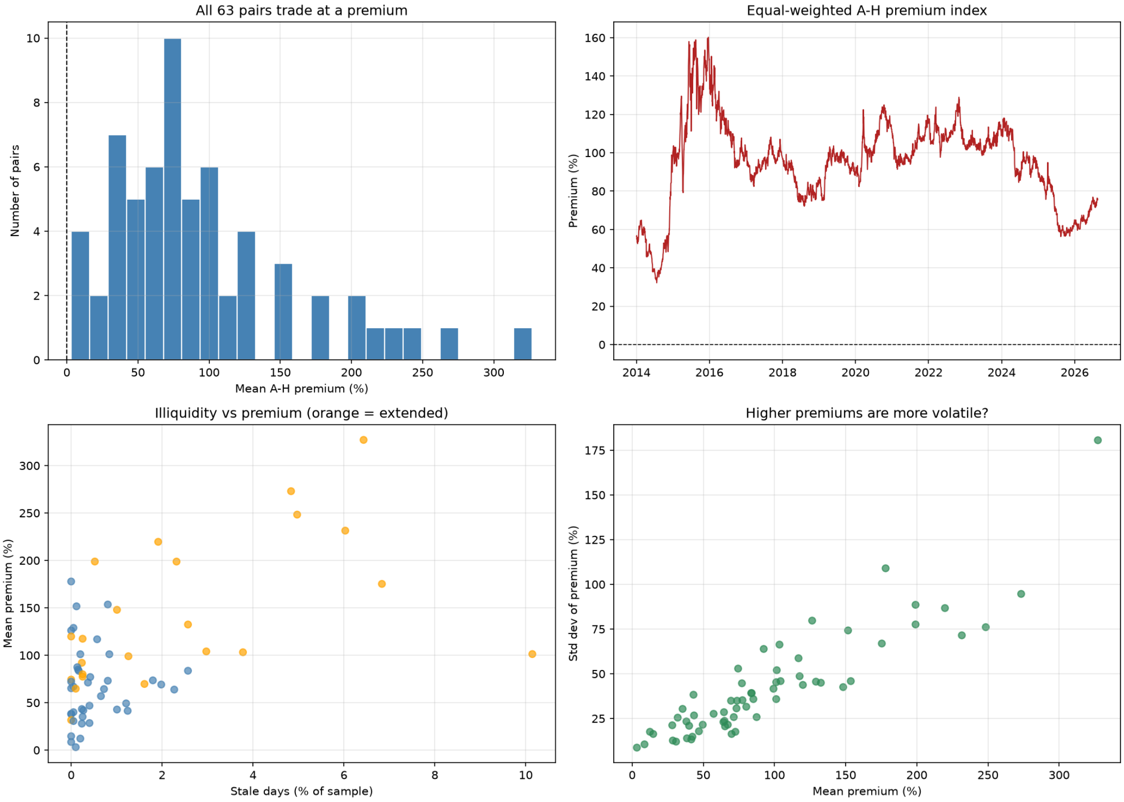

Figure 3: Cross-sectional distribution of mean A-H premiums (top left); equal-weighted premium index over time (top right); illiquidity vs premium by tier (bottom left); mean-vs-volatility relationship (bottom right).

Tier and liquidity. Core (large, liquid) +52.2% vs extended (small, illiquid) +122.8%. However, corr(staleness, premium) = +0.569 pooled but +0.049 *within* core — the relationship is between tiers, not within them, and the pooled figure is confounded by tier membership. Restricted range within core also attenuates the within-group estimate.

Co-movement. Mean pairwise correlation across the 63 premium series is +0.325 (median +0.371; 32% above 0.5; 16% negative). A common factor exists but explains roughly a third of the variation, which motivates pair-by-pair rather than pooled estimation.

6 Stationarity

Tests on log(1 + premium), $n = 2{,}977$ per pair, with Benjamini–Hochberg FDR control (Benjamini and Hochberg, 1995).

Test	Null	Rejections (FDR)	Uncorrected
ADF	unit root	**0 / 63**	21 / 63
Phillips–Perron	unit root	4 / 63	33 / 63
KPSS	stationarity	**63 / 63**	—

The correction is load-bearing: 21 uncorrected ADF rejections become 0. Reporting the uncorrected figure would have produced a materially different — and false — conclusion.

6.1 Robustness to structural breaks

Perron (1989) shows a stationary series with a level shift is systematically misread as a unit root, and we have a documented panel-wide shift (equal-weighted index: 52.1% in 2014 $\to$ 127.1% in 2015).

Zivot–Andrews (Zivot and Andrews, 1992) with an endogenous break in the intercept rejects for **1 / 63**; ADF on the post-2016 subsample rejects for **0 / 63**. Neither allowing a break nor excluding the turbulent period restores stationarity.

Break-date estimates cluster in 2024Q3 (12 pairs) and 2020Q1 (7 pairs), but since the underlying tests reject for only one pair these dates are **suggestive, not identified**: Zivot–Andrews returns an argmin whether or not it rejects.

Break-date estimates cluster near 2024Q3, but this coincides almost exactly with the upper edge of the Zivot–Andrews search window (2015-11-20 to 2024-09-26, after the standard 15% trim on each side of the sample). Since the algorithm cannot search past this date and must still return a candidate break for every pair regardless of whether the null is rejected, this cluster is very likely an artefact of the trimming boundary rather than a genuine concentration of structural breaks, and should not be interpreted further. The 2020Q1 cluster, comfortably interior to the search window, is not subject to the same concern, though with only one pair rejecting the unit root overall (Section 6), neither cluster should be read as strong evidence of a specific break date.

7 Cointegration

7.1 Engle–Granger with free β

Testing the log spread imposes a cointegrating vector of $(1, -1)$. Engle–Granger (Engle and Granger, 1987) relaxes this, estimating $\log P_t^A = \alpha + \beta \log P_t^H + \varepsilon_t$ and testing ε for stationarity with MacKinnon critical values (MacKinnon, 2010), which correct for the fact that OLS has already minimised the residual variance.

Stage	Pairs "cointegrated"
Naive ADF on residuals (*incorrect*)	42 / 63
MacKinnon critical values	15 / 63
+ FDR correction	**0 / 63**

β is not interpretable here. With two $I(1)$ series that are not cointegrated, the regression is spurious (Granger and Newbold, 1974): standard errors are invalid, t-statistics diverge, R^2 is inflated. The finding that $|t| > 1.96$ for $H_0\colon \beta = 1$ in 61/63 pairs is a *symptom* of this, not a result. China Life is the clearest illustration: $\beta = 0.053$, $R^2 = 0.004$.

7.2 Johansen

The trace test rejects rank $= 0$ for 31/63 pairs — contradicting Engle–Granger. A battery of diagnostics, however, shows this figure substantially overstates the evidence for cointegration.

First, the Johansen framework assumes both input price series are $I(1)$; only 50/63 pairs satisfy this prerequisite, so up to 13 of the 31 apparent rejections test a specification the data do not support. Second, of the 31 pairs rejecting rank $= 0$, 30 *also* reject rank $= 1$ — indicating the system has full rank, i.e. both price series already look stationary on their own, which directly contradicts the $I(1)$ assumption the test requires and marks these as artefacts rather than genuine cointegration. Third, the result is highly sensitive to a modelling choice with no obvious correct answer: changing the deterministic specification from a restricted constant

($\texttt{det\_order} = 0$) to none ($\texttt{det\_order} = -1$) drops the rejection count from 31 to 11, and the two specifications agree on only 41/63 pairs.

Requiring all three conditions simultaneously — genuine $I(1)$ inputs, rejection of rank $= 0$, and failure to reject rank $= 1$ — leaves **9 of 63 pairs** as candidates for genuine cointegration, before any correction for multiple testing across those 9. **Conclusion:** the apparent method-dependence between Engle–Granger and Johansen is not a genuine disagreement between two comparably reliable tests. Once diagnosed, the great majority of Johansen's apparent rejections are artefacts of misapplied assumptions or specification sensitivity, and the two methods substantially agree: A–H prices show, at most, weak and isolated evidence of cointegration for a small minority of pairs. This is why the control group in Section 2.3 is necessary rather than decorative.

8 Speed of Adjustment

Model the spread as Ornstein–Uhlenbeck, $dS_t = \theta(\mu - S_t)\,dt + \sigma\,dW_t$, whose exact discretisation is an AR(1) with $\phi = e^{-\theta}$, giving a half-life $h = \ln 2/(-\ln \phi)$.

	A–H (63)	H–ADR (20)
Median ϕ	0.9936	0.0972
Median half-life	107.4 days	0.30 days
Range (half-life)	24.8–390.1 days	0.17–0.56 days

Two cautions. (i) A–H half-lives are *descriptive*. Where the unit-root null is not rejected, h is not a well-identified parameter and its confidence interval extends to infinity. Kendall bias, $\mathbb{E}[\hat{\phi}] - \phi \approx -(1 + 3\phi)/n$, runs *against* the paper's conclusion: true persistence is higher than reported.

(ii) H–ADR $\phi \approx 0.1$ does not mean "reversion in 0.3 days" — it means **no measurable persistence at daily frequency**. Reversion is faster than the sampling interval, and the residual variation is consistent with the non-synchronous closing times discussed in §10. Intraday data would be required to time it.

9 Trading Implementation

The statistical evidence in Sections 6–7.2 argues against, rather than for, an A–H convergence strategy. A trading rule requires a stable, statistically supported equilibrium to trade toward; only 9 of 63 pairs retain even provisional evidence of cointegration after diagnostic screening, and that figure has not been corrected for multiple testing across those 9. Pan et al. (2012), testing a comparable strategy on 16 dual-listed firms with intraday 2007–2009 data, report positive daily profits in only 52.33% of samples — barely above chance, and without correcting for the many pairs and holding windows tested.

More fundamentally, the trade is not implementable; This concern compounds with a separate, purely statistical one. Bailey and López de Prado (2014) show that the expected maximum Sharpe ratio across a set of independent backtests grows systematically with the number of trials attempted, even when the true underlying edge is zero: selecting the single best-performing strategy out of many trials is itself a source of performance inflation, distinct from and additional to transaction costs. With 63 A-H pairs tested, reporting the backtest of whichever pair happened to perform best would repeat precisely the error their Deflated Sharpe Ratio is designed to correct. Computing DSR formally requires the variance of Sharpe ratios across all 63 trials and the moments of the selected strategy's return distribution — inputs this paper's diagnostic screening (Sections 6 –7.2) was not designed to produce, and which are left as a natural extension alongside the ECM noted in Section 10. regardless of the statistical result. Executing a convergence strategy on an A–H pair requires shorting the relatively expensive leg; mainland regulations restrict short-selling of A-shares to a limited, quota-controlled set of eligible securities and impose additional margin and disclosure requirements not present for the H-share leg. Zhang and Zhang (2025) formalise this cost directly, finding that a simulated one-year A–H convergence trade carries a probability of loss exceeding 30% even under favourable assumptions, with pronounced negative skew — a rational arbitrageur may decline the trade even where a statistical edge exists.

Combining a weak and largely non-robust statistical edge, on 14% of the sample at most, with a fundamentally constrained ability to execute the required short leg, this paper does not report a full backtest. The direction of the result is not in serious doubt: any gross edge present in the 9 surviving pairs would need to survive mainland stamp duty, commissions, and the shadow cost of the short-sale constraint, against a base rate — per Pan et al. (2012) — already close to random. A full walk-forward backtest with transaction costs and robustness testing (Deflated Sharpe Ratio, stationary bootstrap, Hansen's SPA test) is identified as future work, conditional on first confirming which of the 9 candidate pairs survive multiple-testing correction.

10 Limitations

- **Non-synchronous closes (H–ADR).** Hong Kong closes at 08:00 GMT, New York at 21:00 GMT — 13 hours and a full US session apart. This is measurement error in the regressor, which attenuates β toward zero. Consistent with this, 18 of 20 estimated βs lie *below* 1 (mean 0.994). The bias inflates measured H–ADR spreads, making the reported contrast **conservative**.

- An error correction model, which would characterise the speed and direction of adjustment toward a long-run equilibrium, was considered but is not reported. Section 7.2 shows that only 9 of 63 pairs retain even provisional evidence of cointegration after diagnostic screening, and this figure has not yet been corrected for multiple testing across those 9. Estimating adjustment dynamics for the full panel would therefore model a relationship the paper's own evidence suggests mostly does not exist; for the small subset of pairs with surviving evidence, an ECM (and Hasbrouck information shares, which decompose price discovery between the two legs) would be a natural extension once that subset is confirmed robust to correction for multiple comparisons.

- **Sample asymmetry.** A–H spans 2014–2026; H–ADR pairs begin 2018–2023, since Hong Kong secondary listings of US-listed Chinese firms only began in November 2019. Cross-sectional comparisons between the two groups (e.g. mean premium, mean half-life) are therefore comparisons across somewhat different sample periods as well as different institutional regimes. A matched post-2021 rerun of the A–H stationarity and cointegration tests, restricting to the period both groups jointly cover, is a natural robustness check not performed here given time constraints; given that the A–H premium was, if anything, larger and more persistent in the post-2021 period than earlier in the sample (Section 5), there is no reason to expect this would weaken the paper's central contrast.

- **Survivorship.** 3 delisted A–H pairs excluded by data availability.

- **Onshore vs offshore CNY.** CNY and CNH diverge under stress; since the object of study is precisely the onshore/offshore wall, the choice of rate is itself an economic assumption.

- **Selection.** Large, liquid, sector-spread names, biasing toward state-owned enterprises. The A–H universe is structurally SOE-heavy because VIE-structured firms cannot list onshore.
- **Ex-dividend jumps** retained under unadjusted prices.

11 Conclusion

Chinese cross-listed equities offer a rare natural experiment in the law of one price: the same set of companies, subject to the same macroeconomic environment, priced in parallel by two different institutional regimes. Where a conversion mechanism connects the two listings — the depositary bank's contractual obligation to create and cancel ADRs on demand — prices are tightly cointegrated: 20 of 20 verified H–ADR pairs, with a cointegrating coefficient within a percentage point of one and R^2 exceeding 0.99. Where no such mechanism exists, and mainland capital controls additionally block any synthetic replication of the trade, cointegration is essentially absent: no A–H pair is cointegrated under Engle–Granger after correcting for multiple testing, and a full diagnostic screen of Johansen's apparently contradictory 31/63 finding leaves at most 9 pairs as provisional candidates, uncorrected for the multiple comparisons across those 9. The premium itself tells the same story at a different scale: -0.15% for H–ADR pairs against $+77.95\%$ for A–H pairs, a gap of two orders of magnitude produced by identical code applied to identical windows.

This finding sits in tension with an earlier literature. Pan et al. (2012) and Cai et al. (2011), testing overlapping samples of A–H firms on data predating late 2014, both report cointegration — though, as Section 3.4 shows, Pan et al. (2012)'s own results are considerably less uniform on closer inspection than their summary suggests, with several firms, including this paper's own ICBC test case, cointegrating on only a minority of the intraday windows they report. Stock Connect launched in November 2014, sitting almost exactly at the boundary between their samples and the one studied here. Two readings of this discontinuity are available, and the evidence in this paper does not fully adjudicate between them. It is possible that Stock Connect itself altered the market's structure, and that the weaker relationship documented here reflects a genuine regime change — consistent with Liu and Zou (2026), Zhang et al. (2022), and Tang and Liu (2026), who each find the connect programs *widened* rather than narrowed the A–H premium, precisely the opposite of the convergence a naive reading of "market integration" would predict. It is equally possible that cointegration in the pre-2014 literature was already closer to the noise floor than commonly summarised, and that the stricter statistical discipline applied here — MacKinnon critical values, false discovery rate correction, and the diagnostic screening of Johansen's own assumptions — would have produced a similarly weak result had it been applied to their data. Both readings converge on the same practical conclusion: cointegration claims for A–H prices are sensitive to sample period, statistical correction, and test specification in a way that a single reported figure obscures, and should be treated with more caution than the existing literature typically affords them.

The broader implication extends beyond this specific market. Froot and Dabora's (1999) evidence from Royal Dutch/Shell shows that even a fully open, fully liquid, unrestricted market does not guarantee price convergence between two claims on identical cash flows — their own price differentials contain a unit root, indistinguishable in that respect from the A–H pairs studied here, despite facing none of A–H's institutional barriers. Read together, the two results suggest that neither the presence of restrictions nor their absence reliably predicts whether two prices will converge. What predicts convergence, at least in the settings studied here, is the presence of a specific, low-friction, close-to-automatic conversion instrument — and its absence, rather than regulation in general, that predicts its failure. The law of one price is not a property of markets left alone, nor a casualty of markets left constrained; it is an artefact of the plumbing connecting them.

A Full pair-level results

The complete pair-level results for all 63 A-H pairs (premium statistics, staleness flags, and jump counts) are available in the project's public GitHub repository at https://github.com/rishabh-verma-stat/dual-listing-spread-dynamics.git. Table 3 summarizes the ten pairs with the largest and smallest mean premiums.

Company	Sector	Mean premium
Top 10		
Luoyang Glass	Materials	327%
Beiren Printing	Industrials	273%
Yizheng Chemical	Materials	231%
Nanjing Panda	Technology	189%
Shandong Xinhua Pharma	Healthcare	178%
China Shipping Container	Shipping	173%
Guangzhou Shipyard	Industrials	165%
Tianjin Capital Env.	Utilities	151%
Shanghai Petrochemical	Materials	128%
Guangzhou Automobile	Autos	125%
Bottom 10		
Bank of China	Banking	27%
Sinopec	Energy	27%
China Construction Bank	Banking	25%
Jiangsu Expressway	Infrastructure	22%
ICBC	Banking	20%
Bank of Communications	Banking	20%
Weichai Power	Industrials	11%
Anhui Conch Cement	Materials	8%
Ping An	Insurance	5%
China Merchants Bank	Banking	4%

Table 3: Ten A-H pairs with the largest and smallest mean A-share premium over the sample period, by company and sector. All 63 pairs exhibit a positive mean premium; "bottom" here denotes the smallest positive premium rather than a discount. Full results for all pairs are in the project repository.

B Data quality report

Table 4 reports per-pair staleness and jump-flag counts across all 63 A-H pairs, sorted by `stale_pct` descending.

Table 4: Staleness and jump diagnostics by A-H pair, sorted by `stale_pct` descending. `stale_a`/`stale_h` count trading days with zero price change (illiquid-quote proxy) on the A- and H-share legs; `jumps` counts log-return outliers flagged at $|z| > 5$; n is the number of aligned trading-day observations; `stale_pct` is $\max(\text{stale\_a}, \text{stale\_h})/n \times 100$.

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Air China	20	20	6	2977	0.67
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Angang Steel	6	9	6	2977	0.25
BYD	7	7	3	2977	0.24
Anhui Expressway	8	6	15	2977	0.24
CITIC Securities	14	0	8	2977	0.24
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Guangzhou Automobile	10	0	8	2977	0.17
Huaneng Power	3	7	5	2977	0.17
PetroChina	9	0	2	2977	0.15
Jiangxi Copper	8	0	5	2977	0.13
China Merchants Bank	3	3	1	2977	0.10
Shenzhen Expressway	0	6	4	2977	0.10
China Pacific Insurance	3	0	3	2977	0.05
China Southern Airlines	3	0	13	2977	0.05
China Comm. Construction	0	3	12	2977	0.05
ICBC	0	3	2	2977	0.05
Huatai Securities	0	0	4	2644	0.00
Tsingtao Brewery	0	0	3	2977	0.00
Guangdong Kelon	0	0	16	2977	0.00
Yanzhou Coal	0	0	21	2977	0.00
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Jiangsu Expressway	0	0	2	2977	0.00
China Life	0	0	5	2977	0.00
Weichai Power	0	0	3	2977	0.00
China Oilfield Services	0	0	5	2977	0.00
Ping An	0	0	1	2977	0.00
CNOOC	0	0	0	1019	0.00

C Reproduction

All code and processed data: https://github.com/rishabh-verma-stat/dual-listing-spread-dynamics.git. All results reported in this paper are computed on data through 20 August 2026. The exact CSVs used to produce every table and figure are committed to the project repository under `data/processed/`, enabling exact reproduction of the reported numbers regardless of future data revisions or market movements.

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